Sahaj Bhadja

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Summary

Software engineer with 2+ years building **high-performance backend systems**, **C++/Python tooling**, and **scalable data workflows**, with strong fundamentals in algorithms, concurrency, and production reliability engineering.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Optimized **multi-tenant backend services** and indexed query execution paths, reducing latency from **1.2s to 350ms** for 500+ users while preserving deterministic behavior under concurrent and data-heavy workloads.
- Implemented **asynchronous FastAPI pipelines** for 5K+ monthly jobs with validation, failure isolation, and reconciliation logic, improving runtime reliability and reducing propagation of errors across dependent service components.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **microservice-based data processing workflows** with Python/Java and AWS Glue/S3, improving response time by **30%** while supporting scalable, resilient execution across distributed production environments.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito, increasing CI pipeline efficiency by **68%** and reducing technical debt by **20%** through continuous verification and defect prevention.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Built high-throughput **backend and full-stack modules** handling 10K+ daily requests, applying structured API design, concurrency-safe processing, and maintainable integration patterns in rapidly evolving product iterations.
- Improved delivery speed via **CI/CD quality gates** and robust regression checks, reducing turnaround time by **60%** while maintaining stable behavior across frequent releases and complex feature dependencies.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Designed **Python analysis pipelines** over 100K+ trajectory logs, generating reproducible diagnostics and benchmarks that improved algorithmic understanding and accelerated engineering decisions on performance-sensitive implementations.
- Produced concise technical reports and debugging summaries for multidisciplinary teams, improving communication of root causes and helping convert exploratory findings into robust, production-oriented engineering tasks.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Developed **REST/WebSocket components** supporting eight workflows and 2M+ synchronized records, improving consistency of real-time state transitions and robustness of user-facing operational data flows.
- Automated Docker/Jenkins deployment and testing pipelines, cutting release time by **50%** and improving reproducibility of build-test-run cycles across Linux-based collaborative development environments.

Skills

Languages: **C++**, **Python**, **Java**, SQL, JavaScript/TypeScript
Core CS: algorithms, data structures, concurrency, object-oriented design, performance optimization
Backend/Systems: FastAPI, Flask, Node.js, microservices, REST APIs, PostgreSQL/MySQL
Infrastructure: Linux, Docker, Kubernetes, Jenkins, Git, CI/CD, observability-focused debugging

Projects

CPU Scheduling Simulator | *React, JavaScript, OS Algorithms*

- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling with interactive Gantt visualization, reducing average turnaround by **25%** and improving debugging efficiency by **20%** across 1K+ process simulations.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built a **schema-aware SQL workspace** across 10+ databases with structured query validation and execution controls, improving correctness and reducing manual troubleshooting in large-scale analytical data operations.

Linguistix | *Python, FastAPI, AWS ECS/Fargate, CodePipeline*

- Developed a containerized API with automated quality checks and CI/CD deployment controls, achieving **87%** processing accuracy while improving reliability and repeatability of cloud-based production execution workflows.